MATH INQUIRERS HOLIDAY PROGRAM

Free 30-minute taster

Lesson 1: Digit Detectives (Numbers to 1000) • Math Inquirers P2 Ages 8-9

Notice it. Test it. Explain why.

What this is

A short sample of a real Math Inquirers lesson for Ages 8-9 (finishing P2). It is Lesson 1 of the Primary 2 holiday program (Digit Detectives), shortened to about 30 minutes: a warm-up, a hands-on explore, a mini-lesson, a few worksheet questions and an exit ticket. The same routine runs through all ten 90-minute lessons.

The 30 minutes

Time	Part	What happens
5 min	**Warm-up**	Place It Right: the tutor draws digit cards one at a time. After each card, children must write it in the hundreds, tens or ones box straight away and cannot move it. Biggest number wins. Then play again for the smallest.
8 min	**Explore**	Pairs take the digit cards 3, 5 and 8 and make every 3-digit number they can, using each card once. They record each number, then find the biggest and the smallest and decide how they know they have found them all.
5 min	**Mini-lesson**	Place value: 5 hundreds, 12 tens and 4 ones is 500 + 120 + 4 = 624. Make a list in order: fix the hundreds digit, then swap the other two (1, 6, 9 gives 169, 196, 619, 691, 916, 961). Biggest: biggest digit in the hundreds place. Zero can never be the hundreds digit. Before page 2: on a 1000 chart one square right is 1 more, one square down is 10 more; count on and back across a hundred (608, 598, 588); solve a digit riddle by listing the tens and ones that fit.
9 min	**Worksheet**	The child works on questions 1 to 4 of the Lesson 1 sheet (next pages), then checks them with the tutor.
3 min	**Exit ticket**	Use the digit cards 6, 0 and 2 once each. Write the biggest and the smallest 3-digit number.

What you need

Digit cards 0 to 9 (one set per child), place-value mat (hundreds, tens, ones), base-ten blocks or paper strips and squares, mini whiteboards, a 1000 chart page (rows of 10).

Online: Children cut out paper digit cards before the lesson and move them on camera; the tutor writes on the slide 1000-chart pieces.

Pages in this file: this page, the child's worksheet (about two pages), then tutor notes with the answers. Do not give the tutor-notes pages to the child. This taster is enrichment for thinking skills; it does not promise exam results and is not affiliated with MOE or the Gifted Education Programme.

DIGIT DETECTIVES

Name _____ Group _____ Date _____

A 3-digit number has hundreds, tens and ones. Where a digit sits tells you what it is worth.

No.1

Place value. Fill in the blanks.

a 4 hundreds, 0 tens and 7 ones = _____

b 6 hundreds, 15 tens and 2 ones = _____

c 830 = 8 hundreds and _____ tens 830 = 7 hundreds and _____ tens

d In 572, the digit 7 is worth _____

No.2

Use the digit cards 2, 7 and 4. Use each card once to make a 3-digit number.



a Make a list in order. Write every number you can make.

Start with 2	Start with 4	Start with 7
_____ _____	_____ _____	_____ _____

b Biggest number: _____ Smallest number: _____

c Which of your numbers is closest to 500? _____

d How many of your numbers are odd? _____

No.3

Now use the digit cards 0, 5 and 9.



a List every 3-digit number you can make, in order.

Numbers: _____

b The cards 2, 7, 4 made 6 numbers. Why do 0, 5, 9 make fewer?

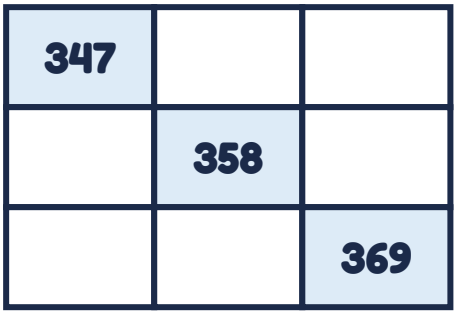
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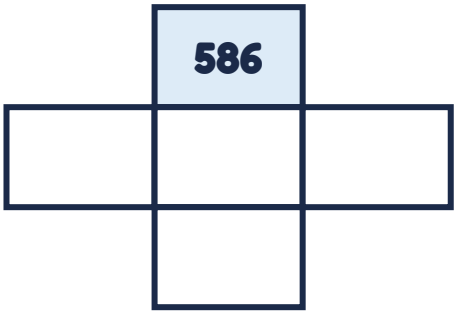
No.4

These are pieces cut from a 1000 chart. Each row has 10 numbers. Fill in the empty squares.

Piece A



Piece B



a On the chart, one square DOWN is _____ more. One square RIGHT is _____ more.

b Which number is 2 squares below 385 on the chart? _____

Tutor notes: Lesson 1, Digit Detectives

Numbers to 1000 • Ages 8-9 (finishing P2)

Goal

Read and build 3-digit numbers by place value (also with more than 9 tens), make every number from a set of digit cards by listing in order, find the biggest and smallest, and use patterns on a 1000 chart to count on and back in 1s, 10s and 100s.

Skills: Place value, making a list in order, looking for a pattern, reasoning about zero

Opening

Welcome, Digit Detectives! Every 3-digit number is made of just three digits, but WHERE a digit sits changes everything.

Today we find every number hiding in a set of digit cards, and we make sure we do not miss any. Notice it. Test it. Explain why.

Warm-up (about 5 minutes)

Place It Right: the tutor draws digit cards one at a time. After each card, children must write it in the hundreds, tens or ones box straight away and cannot move it. Biggest number wins. Then play again for the smallest.

Draw three boxes on your whiteboard: hundreds, tens, ones. I will turn over one digit card at a time. As soon as you see it, write it in a box. No moving it later!

Try to make the BIGGEST number. Where will you put a 9? Where will you put a 1? Why?

Explore (about 8 minutes)

Pairs take the digit cards 3, 5 and 8 and make every 3-digit number they can, using each card once. They record each number, then find the biggest and the smallest and decide how they know they have found them all.

Predict: You have the cards 3, 5 and 8. How many different 3-digit numbers do you think you can make with them? Write your guess in the corner of your whiteboard.

- Stuck: put the 3 in the hundreds place first. What can the other two cards do?
- Not sure if you have them all: “How many numbers start with 3? With 5? With 8?”
- Finished early: add a 0 card and make numbers with 0, 3 and 5. Why are there fewer?

Mini-lesson (about 5 minutes)

Place value: 5 hundreds, 12 tens and 4 ones is 500 + 120 + 4 = 624. Make a list in order: fix the hundreds digit, then swap the other two (1, 6, 9 gives 169, 196, 619, 691, 916, 961). Biggest: biggest digit in the hundreds place. Zero can never be the hundreds digit. Before page 2: on a 1000 chart one square right is 1 more, one square down is 10 more; count on and back across a hundred (608, 598, 588); solve a digit riddle by listing the tens and ones that fit.

Worksheet answers (questions 1 to 4)

Q1	(a) 407 (b) 752 (c) 3 tens; 13 tens (d) 70
Q2	(a) Start with 2: 247, 274. Start with 4: 427, 472. Start with 7: 724, 742. (b) Biggest 742, smallest 247 (c) 472 (only 28 away from 500) (d) 2 odd numbers: 247 and 427
Q3	(a) 509, 590, 905, 950 (b) 0 cannot be the hundreds digit (059 is not a 3-digit number), so only 5 or 9 can start: 2 + 2 = 4 numbers, not 6.
Q4	Piece A: 348, 349 / 357, 359 / 367, 368. Piece B: 595, 596, 597 in the middle row and 606 at the bottom. (a) Down: 10 more. Right: 1 more. (b) 405 (395, then 405)

Talk time

Which is bigger, 401 or 399? Ali says 399 because it has two 9s. Who is right, and why?

Answer: 401 is bigger. Compare the hundreds first: 4 hundreds is more than 3 hundreds, so the tens and ones do not matter. We know we have every number because we listed them in order: each card starts the same number of numbers.

Exit ticket (about 3 minutes)

Use the digit cards 6, 0 and 2 once each. Write the biggest and the smallest 3-digit number.

Answer: Biggest 620, smallest 206.

Watch for these mistakes

- Writing 6 hundreds, 15 tens and 2 ones as 6152 instead of regrouping 15 tens as 1 hundred and 5 tens (752).
- Listing digit-card numbers at random, so one is missed or written twice, or counting 059 as a 3-digit number.
- Counting back in 10s across a hundred: 512, 502, 592 instead of 512, 502, 492.

Extension for fast finishers

Use any three of the cards 1, 3, 5 and 7. How many different 3-digit numbers can you make? (24: 6 for each set of three cards, and there are 4 sets.)

After the taster

Tell the parent one thing their child noticed and one thing they explained well. Then share the next steps: [dates, fee and sign-up link]. The free diagnostic for the matching level helps choose between Ages 8-9 (finishing P2) and the level either side (see the placement note).